# 📉 WhyDrop

The agent that can explain *why* sales dropped, and proves every claim.

GenAI Works Open Agent Hackathon 2026 · Track 2 · Nathan Melia · github.com/jamesbrowndevs/whydrop

# Problem & users

- Small online sellers (Gumroad, Shopify, Etsy) watch revenue fall with no analyst to ask.
- Dashboards show **what** dropped, not **why**.
- The clues are scattered: order exports, ad spend, traffic, refunds DB, price and release logs.
- Generic chatbots guess, and an unverified "why" leads to wrong fixes.

# Solution

Ask in plain English: *"Why did revenue drop in September?"*

- Ranked causes with confidence and estimated $ impact
- Every claim cites exact cells (query · row · column · value)
- Rejected hypotheses shown with their evidence
- A recommended action per cause; Markdown report export

# How it works

① Signal scan on Zetaris → ② Plan hypotheses + recall memory → ③ SQL per hypothesis

④ Analyse → cited claims → ⑤ Verify every claim → ⑥ Ranked report → ⑦ Remember (H-MEM)

Hypotheses are tested in parallel; the UI streams every step, query and model call live.

# Architecture

| Layer | Tech | Role |
|---|---|---|
| Data | Zetaris Lightning (Spark) | WHYDROP_FILES (CSV/JSON filestore) + WHYDROP_PG (Postgres JDBC) federated; the agent's only tool is zetaris_sql() |
| Reasoning | NVIDIA Nemotron | planner & analyst: 3 Super 120B · SQL: 3 Nano reasoning · verifier: 3.5 Lightning, with routing + fallback |
| Memory | Meterless H-MEM | verified causes and rejections stored with provenance and recalled next run |
| UI | Streamlit | live steps, SQL + rows, routing, ranked report |

# Trust: no claim without evidence

1. The cited query is re-run on Zetaris, and the result hash must reproduce
2. Each cited cell must exist and match
3. Every number in the text must come from cited cells (or a simple delta, ratio or %)
4. An independent Nemotron verifier lane must agree

Failing claims are struck out and never reach the answer.

# Sponsor tech

- **Zetaris**: one SQL layer over files + Postgres; cross-source joins; verification re-runs
- **Meterless**: H-MEM long-term memory of investigations (agentathon starter engine)
- **NVIDIA Nemotron**: four lanes, routing visible in UI, logs and reports
- **Cursor**: development environment

# Results

Synthetic seller dataset (Jun–Sep 2026) with 3 planted causes + 2 red herrings; the agent never sees the ground truth. Only verified claims are scored.

| Question | Result | Causes | Rejected | Time |
|---|---|---|---|---|
| Why did revenue drop in September 2026? | ❌ fail | 5 | 1 | 277s |
| Is our ad spend still working? | ✅ pass | 3 | 3 | 560s |
| Why did Notion Freelance OS sales fall even though we raised the price? | ✅ pass | 4 | 2 | 420s |
| Why are refunds up in the second half of September? | ✅ pass | 1 | 5 | 538s |
| Did email marketing cause the drop? | ✅ pass | 4 | 3 | 583s |

**4/5 passed** · 2/3 planted causes found on the headline question · run 2026-10-24 01:52 BST

# What's unique

- Explains *and* rejects: shows the hypotheses it ruled out, with proof
- Deterministic + LLM verification of every number
- Federated data without ETL, via Zetaris
- Learns across investigations with Meterless memory

# Future work

- Live Shopify / Gumroad / Stripe connectors
- Watchdog mode: alert on a drop with the explanation attached
- Causal-impact estimates; hosted multi-tenant version

github.com/jamesbrowndevs/whydrop